Electromagnetic Theory & Quantum Physics: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

15 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10
11 12 13 14 15

Radiation and Quanta: worked solutions

Solution 1 — Count the packets. Return →

$f = c/\lambda = \mathbf{5.996 \times 10^{14}\ Hz}$. $E = 1239.84/500 = \mathbf{2.480\ eV} = 3.973 \times 10^{-19}$ J. A joule per second divided by joules per photon gives 2.52×10^{18} **photons/s**.

Check. Higher power at the same colour means more photons, not more energy per photon.

Solution 2 — Ten parsecs away. Return →

$F = 4L_\odot/[4\pi(10\,\mathrm{pc})^2]$. With $d = 3.08568 \times 10^{17}$ m, $F = 1.28 \times 10^{-9}$ W m^{-2}. This is bolometric flux, not a visual-band measurement.

Check. Doubling the distance would reduce the flux fourfold.

Solution 3 — A green-ish peak. Return →

$T = 2.89777 \times 10^{-3} / (500 \times 10^{-9}) = \mathbf{5796\ K}$. The surface flux is $\sigma T^4 = \mathbf{6.40 \times 10^7\ W\ m^{-2}}$. A blackbody emits a broad range of wavelengths; human colour perception integrates that range. A peak in green does not make a star monochromatic green.

Check. This temperature is close to the Sun's, whose integrated visible light is approximately white.

Solution 4 — Infer a radius. Return →

Use $L \propto R^2 T^4$. Thus $R/R_\odot = \sqrt{9/2^4} = \sqrt{9/16} = \mathbf{0.75}$. The hotter surface compensates for the smaller area.

Check. $(0.75)^2 (2)^4 = 9$.

Solution 5 — Same temperature, different size. Return →

At equal temperature, flux ratio equals area ratio, $2^2 = 4$. Therefore $\Delta m = -2.5 \log_{10} 4 = \mathbf{-1.505}$. The larger star has the smaller magnitude.

Check. Brighter means more negative magnitude difference.

Solution 6 — The stopping voltage. Return →

$\phi = 1239.84/400 = \mathbf{3.100\ eV}$. At 250 nm, $E_\gamma = 4.959$ eV, so $K_{\max} = \mathbf{1.860\ eV}$ and $V_{\text{stop}} = \mathbf{1.860\ V}$. Doubling intensity doubles the available photon rate and, in the unsaturated ideal case, the emission rate; it does not change $K_{\max}$.

Check. The incident photon energy exceeds the threshold.

Solution 7 — A photon budget. Return →

The collecting area is $\pi (1)^2 = 3.1416$ m². Counts $N = 1000 \times 3.1416 \times 0.25 \times 60 = \mathbf{4.71 \times 10^4}$; ideal SNR $= \sqrt{N} = \mathbf{217}$. A 4 m telescope has four times the area, so SNR becomes **434**.

Check. SNR grows with diameter, not diameter squared, when photon shot noise dominates.

Solution 8 — A solar sail. Return →

$p_{\text{abs}} = F/c = \mathbf{4.54 \times 10^{-6}\ Pa}$; perfect reflection gives 9.08 μ**Pa**. On 100 m² the absorbing force is $pA = \mathbf{4.54 \times 10^{-4}\ N}$.

Check. Radiation pressure is small because c is large. Oblique incidence changes the projected area and momentum direction.

Solution 9 — Which peak? Return →

$B_\lambda d\lambda$ and $B_\nu d\nu$ describe the same power in corresponding intervals, so $B_\lambda = B_\nu |d\nu/d\lambda| = B_\nu c/\lambda^2$. The extra Jacobian changes the shape and shifts the maximum. **Integrated power over corresponding intervals is conserved**, not the numerical location of a density maximum.

Check. Always state whether a peak is per unit wavelength or per unit frequency.

Solution 10 — Compensating changes. Return →

$L'/L = (1/2)^2(2)^4 = \mathbf{4}$: the luminosity quadruples, not stays fixed. Wien's law gives $\lambda'_{\max}/\lambda_{\max} = 1/2$. Radius affects total area, not the peak wavelength.

Check. Surface flux grows sixteenfold while area falls fourfold.

Solution 11 — From angle to temperature. Return →

Idea. Convert angular diameter to physical diameter before using the luminosity law. At 100 pc, $0.002''$ corresponds to diameter $0.002 \times 100 = 0.2$ AU, so $R = 0.1$ AU $= \mathbf{21.50 R_\odot}$. Then

$$T = 5772 \left(\frac{100}{21.50^2}\right)^{1/4} = \mathbf{3936\ K}. \tag{1}$$

Check. This is a large, cool luminous star. Using the angular diameter as an angular radius would double the radius and lower the inferred temperature incorrectly.

Solution 12 — Do not saturate. Return →

Electron rate $= 0.8 \times 10^5 = 8.0 \times 10^4$ s^{-1}. Thus $t_{\max} = 50000/80000 = \mathbf{0.625\ s}$. Short exposures avoid full-well saturation; stacking can restore total counts and improve SNR, though read noise and overheads must be included in a real instrument.

Check. The limit is set by electrons stored, not photons incident.

Solution 13 — A moving photon source. Return →

For $v/c = 0.001$, $\Delta\lambda/\lambda \simeq +0.001$ and $\Delta E/E \simeq -0.001$: **0.1% redder and 0.1% less energetic per photon**. The photon still travels at c in vacuum. The energy difference is frame-dependent, not a slowing photon.

Check. The exact relativistic shift differs only at order $(v/c)^2$ here.

Solution 14 — Dust at the blowout scale. Return →

Substitution in $\beta = 3L/(16\pi cGM\rho s)$ gives $\beta = 0.287$. Since $\beta \propto 1/s$, the radius for $\beta = 0.5$ is $1\,\mu\text{m} \times (0.287/0.5) = \mathbf{0.574}\ \boldsymbol{\mu}\mathbf{m}$.

Check. Smaller grains feel more radiation force per unit mass. Actual blowout also depends on the grain's release orbit and optical properties.

Solution 15 — An eclipse in two bands. Return →

Idea. An eclipse removes the planet's light. Bolometrically, the contrast is area ratio times T^4 ratio:

$$\delta_{\text{bol}} = (0.1)^2(1500/6000)^4 = \mathbf{3.91} \times \mathbf{10^{-5}} = 39.1\ \text{ppm}. \tag{2}$$

At the same wavelength, the Planck prefactor cancels. With $x = 14387.77/(\lambda_{\mu\text{m}} T)$,

$$\delta_{10\mu\text{m}} = 0.01\frac{e^{x_*} - 1}{e^{x_p} - 1} = \mathbf{1.68} \times \mathbf{10^{-3}} = 1684\ \text{ppm}. \tag{3}$$

Check. In the infrared the cool planet is much less disadvantaged relative to the star than in total power. A real bandpass requires integration over the filter and stellar/planetary spectra.